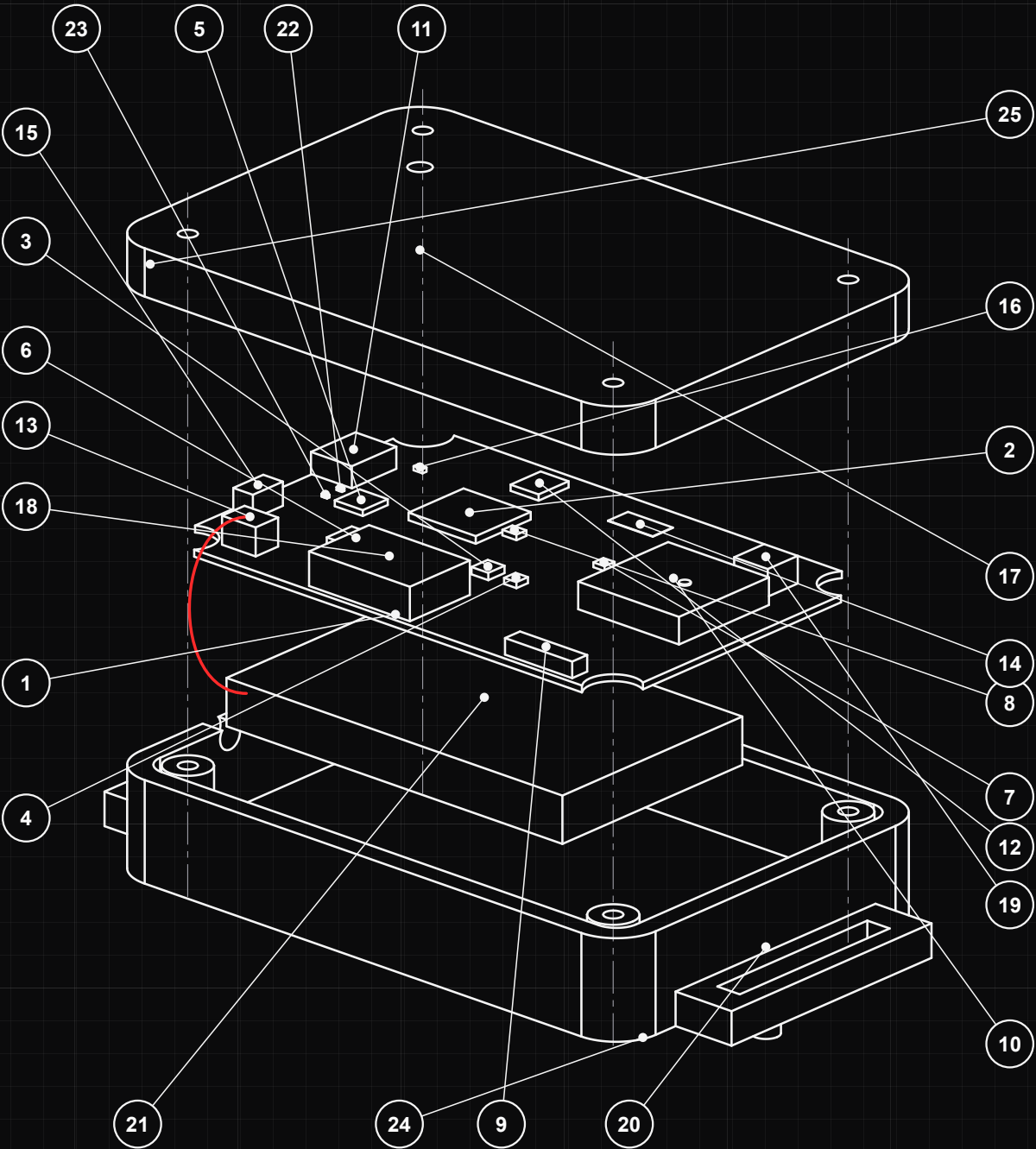




EXPLODED ISOMETRIC / 25 CALLOUTS / SCREW AXES IN CHAIN LINE / W1 BATTERY CABLE IN RED

PARTS LIST

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

NO	QTY	DESIGNATION	PART	SIZE mm	DATA
1	1	PCB1 Main board RL-MB-01	Redline design	80 x 52	design choice
2	1	U1 nRF9151 SiP- LTE-M, NB-IoT, NTN, DECT NR+, GNSS	Nordic nRF9151	12 x 11	maker (height indicative)
3	1	U2 ADXL372 shock sensor, +-200 g	Analog Devices ADXL372	3 x 3.25	maker
4	1	U3 ADXL367 motion sensor, +-2/4/8 g	Analog Devices ADXL367	2.2 x 2.3	maker
5	1	U4 nPM1300 PMIC: charger, fuel gauge, regulators	Nordic nPM1300 QFN32	5 x 5	maker (height indicative)
6	1	U5 64 Mbit low-power flash: the black box memory	Macronix MX25R6435F	6 x 5	indicative (WSON-8)
7	1	U6 GNSS filter then LNA	to choose	2 x 2	indicative
8	1	U7 Hold-up charger and ideal diode for C1	to choose	2.5 x 2	indicative
9	1	ANT1 Chip antenna 824-960 / 1710-2690 MHz	Ignion NN02-224	12 x 3	distributor
10	1	ANT2 GNSS patch antenna 18 mm	Taoglas CGGBP.18.4.A.02	18 x 18	maker
11	1	J1 USB-C receptacle (charge and black box readout)	to choose	7.3 x 8.9	indicative
12	1	J2 eSIM, soldered MFF2	to choose (IoT operator)	5 x 6	indicative
13	1	J3 Battery connector JST-PH 2, SMD	to choose	6 x 4.5	indicative
14	1	J4 SWD pads Tag-Connect TC2030	footprint only	8 x 4	footprint
15	1	SW1 Side button: arm, disarm, pair	to choose	3.5 x 6	indicative
16	1	D1 Status LED 0603	to choose	1.6 x 0.8	indicative
17	1	LP1 Light pipe PMMA 3 mm	to choose	3 x 3	indicative
18	1	C1 Hold-up supercapacitor	to size	18 x 12	indicative
19	1	TS1 Tamper switch (mounting)	to choose	6 x 6	indicative
20	1	PL1 Tamper plunger, 4 mm, spring loaded	to choose	4 x 4	indicative
21	1	BT1 Li-Po 1S 2000 mAh, protected	Adafruit 2011	60 x 36	distributor
22	1	R1 CC1 pull-down 5.1 kOhm	0402	1 x 0.5	indicative
23	1	R2 CC2 pull-down 5.1 kOhm	0402	1 x 0.5	indicative
24	1	Base with strap ears	RL-HW-01	112 x 62	CAD model
25	1	Lid (non-metallic polymer)	RL-HW-02	92 x 62	CAD model
26	4	Plastic self-tapping screw	to choose	D2.7	to define
27	1	Adhesive foam under the battery	to choose	60 x 36 x 0.5	to define
28	1	Strap or mounting plate	to choose	25 wide	to define

Sources: hardware/netlist.json. "indicative": size to be taken from the part chosen.

- 92 x 62 x 22 mm (112 mm with the strap ears)
- 23 electronic items, one cable inside
- Cellular, satellite and DECT NR+ mesh from one SiP
- Shock +-200 g, motion, tamper plunger, black box flash
- Hold-up capacitor for the last message

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

EXPLODED VIEW: EVERY COMPONENT

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-HW-03	SCALE 1.15:1 (exploded)	REV 0	SHEET 3 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	